Sahil Mukesh Deo

B.Tech, Computer Science and Engineering
Indian Institute of Technology Bombay
Expected graduation: May 2029

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Examination	Institute	Year	CPI/%
Graduation	Indian Institute of Technology Bombay	2025 – present	9.68
Intermediate	Narayana e-Techno School	2025	96.8%
Matriculation	Wisdom High International School	2023	99.2%

Pursuing a **Minor** in **Statistics**
Research interests: randomised algorithms, probabilistic method, derandomisation, complexity classes

Scholastic Achievements

- Received the **Institute Academic Award** for securing **Rank 25** among 1,360 students by CPI *2026*
- Secured **Department Rank 15** among 200 students in CSE, IIT Bombay, by CPI *2026*
- Secured **All India Rank 7** in **JEE Advanced** among 180,000+ candidates *2025*
- Achieved **All India Rank 102** in **JEE Main** among 1,400,000+ candidates *2025*
- Awarded the **Gold Medal** in the **Homi Bhabha Balvaidnyanik Competition** *2019-20*
- Secured **State Rank 5** in the **Maharashtra Scholarship Exam** among 200,000+ students *2018-19*
- Shortlisted for the **Aditya Birla Scholarship** interview *2026*

Olympiads & Competitions

- Among the **top 300** of 40,000+ candidates in the National Standard Examination in Physics (NSEP), selected for the **Indian National Physics Olympiad (INPhO)**, HBCSE *2024-25*
- Cleared the National Standard Examination in Astronomy (NSEA) among 40,000+ candidates to qualify for the **Indian National Astronomy Olympiad (INAO)**, HBCSE *2024-25*
- Cleared the National Standard Examination in Chemistry (NSEC) among 40,000+ candidates to qualify for the **Indian National Chemistry Olympiad (INChO)**, HBCSE *2024-25*
- Secured **1st position** in **Physics Bazinga**, Maths & Physics Club, IIT Bombay *2025*
- Secured **1st position** in **Maths Bazinga**, Maths & Physics Club, IIT Bombay *2025*
- **3rd** in the quizzing round of **The Graviton Crucible**, IIT Bombay Edition *2026*
- Among the **top 600** of 10,000+ participants to reach the in-person regionals of **IICPC QuantFest**, and secured **Rank 227** of 600 at the regionals *2026*

Key Project | Research-Oriented

Randomised Algorithms: A 137-Page Monograph | *Summer of Science* *(2026)*
Maths and Physics Club, IIT Bombay *GitHub: Sahil-M-Deo/Randomised-Algorithms*

- Authored a self-contained **137-page, 13-chapter book** in LaTeX with **6 appendices** (probability, concentration, limit theorems, finite fields, linear algebra) and **original problem sets**
- Covered **RP, ZPP & BPP**, analysis of QuickSort and QuickSelect, **universal & k-wise independent hashing**, **Freivalds' algorithm**, skip lists, treaps, Bloom filters, cuckoo hashing, **Miller–Rabin**, **Pollard's rho**, streaming (**Flajolet–Martin, HyperLogLog**), zero-knowledge proofs and **derandomisation**
- Using the **probabilistic method**, designed Las Vegas algorithms for dominating set and max-cut; proved a success probability of at least $\frac{1}{m+1}$ for finding a cut with $\geq m/2$ edges, giving $\Theta(nm)$ expected time
- **Conjectured** from large-scale simulation (exhaustive for small n, Monte Carlo for large n) that the **5-cycle is the worst-case graph** for the randomised **max-cut** algorithm (success probability $\frac{5}{16}$), which would tighten the bound to $\Theta(n+m)$ expected time
- Proved via a **rank argument** on ± 1 orthogonal columns that n pairwise-independent bits need $\geq \log_2(n+1)$ truly random bits, tight for the **XOR construction**; derandomised max-cut via **conditional expectations**
- Validated an **approximate-independence** assumption with **Monte Carlo sampling** (10,000 trials per n), and showed the **union bound** is near-tight in practice

Professional Experience

First-Year Trading and Technology Program (FTTP) | *Jane Street* *(2026)*
Hong Kong

- Shortlisted for **Jane Street's** selective program for first-year STEM undergraduates
- Attended classes on **trading and technology** and competed in a **team-based mock trading simulation**
- Developed **fair-value estimates** for a complex **live market simulation** with market inefficiencies

Other Projects

Deep Chess | *Seasons of Code*
Web and Coding Club, IIT Bombay

- Built a C++ chess engine rated **2200**, benchmarked against **Stockfish**, with **alpha-beta search**, iterative deepening, **quiescence search**, a **Zobrist-hashed transposition table** and killer-move ordering
- Trained an **NNUE evaluation network** in **PyTorch** with incremental accumulator updates for fast inference, benchmarked against the classical evaluation in **cutechess** tournaments

Game Theory | *Learners' Space*
Maths and Physics Club, IIT Bombay

- Proved **Zermelo's theorem** for infinite-play "Long Chess", and by **induction** that each combinatorial-game position lies in exactly one outcome class $(\mathcal{L}, \mathcal{R}, \mathcal{N}, \mathcal{P})$, set by which player can force a win
- Applied **iterated elimination** of strictly and weakly dominated strategies, pure **Nash equilibria** and **maxmin** security levels; solved the blue-eyed islanders **common-knowledge** puzzle by induction
- Studied **mixed equilibria**, the **minimax theorem** (applied in the monograph) and **Nash's theorem** via **Brouwer's fixed-point theorem**; used the **indifference principle** and solved zero-sum games as **LPs**
- Showed that in the n-player **volunteer's dilemma**, P(no one volunteers) tends to the **cost/benefit** ratio, not 0; analysed a lowest-unique-number **lottery game**; averaging **97.33%** marks on assignments

Reinforcement Learning for Algorithmic Trading | *Winter in Data Science*
Analytics Club, IIT Bombay

- Compared **greedy, ε-greedy and UCB** on multi-armed bandits and modelled environments as **MDPs**
- Built a **Q-Learning** stock trading agent on historical prices, evaluated against a **buy-and-hold** baseline

Mini Game Hub | *Course Project: Software Systems Lab (CS108)*

- Built a two-player hub in **Bash** and **Python/pygame** with three games, time controls and leaderboards
- Blocked **injection** in the **Bash login** by reading input raw so backslashes stay literal, allowing only letters, digits and underscores in usernames, and matching **SHA-256** hashes exactly; three failures lock the account
- Found an **option-injection** bug in `echo -n "$@"` and documented all Bash login risks in the report

Positions of Responsibility

- **Owner and maintainer** of an open-source course resource repository (past papers, reference books) used by **200 batchmates** and **freely available** to juniors and learners outside IIT Bombay *2026*
- Conducted a **mid-semester revision tutorial** for **Discrete Structures (CS105)** under the **Tutorial Services Centre**, creating **original practice problems** and resolving students' doubts *2026*

Technical Skills & Courses

Languages	C++, Python, Java, JavaScript, Bash, Verilog, Assembly, LaTeX
Tools	Git, Make, Pygame, NumPy, Z3 Solver (CNF-SAT)
Competitive Programming	Rated **1654** on Codeforces
Courses	Computer Programming and Utilisation (CS101), Discrete Structures (CS105), Software Systems Lab (CS108)
Ongoing Courses	Data Structures & Algorithms (CS213)†, Digital Logic Design & Computer Architecture (CS230)†, Logic for Computer Science (CS228), Data Analysis & Interpretation (CS215)

†*with corresponding lab (CS293, CS231)*